

B.Sc. Mathematics & Spatial Vectors - Marking Scheme

Q1 (4 marks) Define a vector space and give one example.

Q2 (6 marks) Prove that the dot product is commutative.

Q3 (5 marks) Find the cross product of $(1,2,3)$ and $(4,5,6)$.

Q4 (5 marks) Explain linear independence of vectors.

Q5 (6 marks) Compute the angle between two given vectors.

Q6 (4 marks) State the triangle inequality with a short proof.

Total: 30 marks